

Binary Search

target = 20

5	7	10	15	20	30	33	40	55	61
0	1	2	3	4	5	6	7	8	10

target = "ship"

ln

sat

Search space/interval

target 20

$$\textcircled{1} \left[\begin{array}{l} \underline{l=0} \quad \quad \quad \underline{h=10} \quad \quad \quad \underline{\text{mid} = \frac{(0+10)}{2} = 5} \end{array} \right]^{(n-1)}$$

$$\text{key} = \text{arr}[\text{mid}] = 30$$

key $\neq$ target

0	1	2	3	4	5	6	7	8	9	10
						30				

target < key

target > key

0 - 4

$$h = \text{mid} - 1$$

$$\Rightarrow l=0, h=4$$

② $l=0$ $h=4$ $mid = \left(\frac{0+4}{2} \right) = 2$

$key = arr[mid] = arr[2] = 10$

$target = 20$

$key \neq target$ $(10 < 20)$

$\downarrow$ $\downarrow$

$target < key$ $key < target$

$0 - 4$

$\quad \quad \quad \downarrow$

$\quad \quad \quad \times$

$\quad \quad \quad [3, 4]$

$l = mid + 1$

$l=3 \quad h=4$

Maths.floor()

③ $l=3$ $h=4$ $mid = \left(\frac{3+4}{2} \right) = \lfloor 3.5 \rfloor = 3$

$key = arr[mid] = arr[3] = 15 \quad | \quad target = 20$

$key < target$

$l = mid + 1$

$l=4 \quad h=4$

$$\textcircled{4} \quad l = 4 \quad h = 4 \quad \text{mid} = \left(\frac{4+4}{2} \right) = 4$$

$$\text{key} = \text{arr}[4] = 20$$

$$\boxed{\text{key} == \text{target}}$$

mid → answer

$$\boxed{\text{Assume arr}[4] = 22}$$

$$l = 4$$

$$h = 4$$

$$\text{mid} = 4$$

$$\text{key} = \text{arr}[4] = 22$$

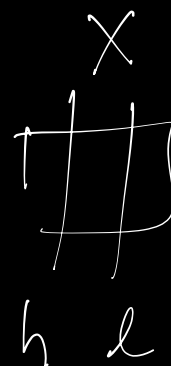
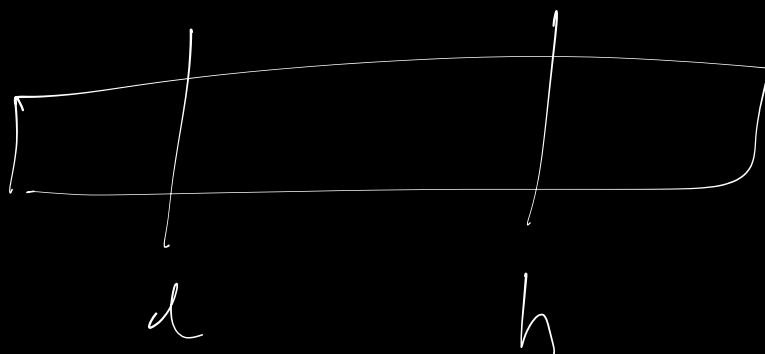
$$\text{target} < \text{key}$$

$$h = \text{mid} - 1$$

$$= 4 - 1$$

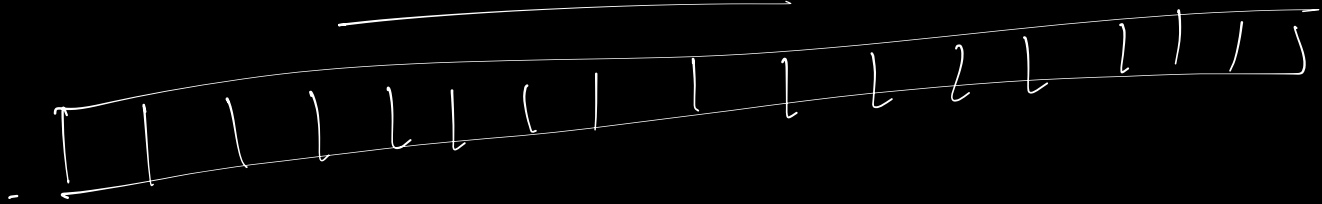
$$l = 4$$

$$h = 3$$



$l > h \Rightarrow$ we should stop

return -1



(1) (2) (3) (4) ... (k)

$$f = \underbrace{1 + 1 + 1 + \dots + 1}_k$$

$$= k \times 1$$

$$= \underline{\underline{k}}$$

K in terms of n



HW

Assign



4th M

4th March 2026

1	7	20	21	30	40	62	70
0	1	2	3	4	5	6	7

$$\text{target} = 62 \checkmark$$

$$\text{target} = \underline{\underline{80}}$$

$$\text{target} = 40$$

$$\text{target} = 30$$

floor values

$$\underline{\text{target} = 62}$$

$$l = 0$$

$$h = 7$$

$$\text{mid} = \left(\frac{0+7}{2} \right) = \underline{\underline{3}}$$

$$\text{key} = \text{arr}(\text{mid}) = \text{arr}(3) = 21 \neq 62$$

key $\rightarrow$ t
0

$$\underline{l = \text{mid} + 1}$$

$$l = 4$$

$$l = 4$$

$$h = 7$$

$$\text{mid} = \frac{4+7}{2} = \underline{\underline{5}} = 5$$

$$\text{key} = \text{arr}(5) = 70 < 62$$

$$l = \text{mid} + 1 = 6$$

$$l = 6$$

$$h = 7$$

$$\text{mid} = 6$$

$$\text{key} = \text{arr}(6) = 62 = 62 \checkmark$$

$$\boxed{\text{ans} = 6}$$

$$\begin{array}{c} \downarrow \\ \begin{array}{c} 0 \qquad \qquad \qquad 7 \\ \boxed{1 \mid 7 \mid 20 \mid 21 \mid 30 \mid 40 \mid 62 \mid 70} \\ 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \end{array} \end{array}$$

target = 80

$$\lfloor 7.5 \rfloor = 7$$

$$\begin{aligned} \text{mid} &= \frac{l+h}{2} \\ &= \frac{0+7}{2} = 3 \end{aligned}$$

Search space
↓
Range for our possible answer

$$l = 0$$

$$\begin{aligned} h &= n-1 \\ &= 7 \end{aligned}$$

(key) → (target)

$$\text{key} = \text{arr}[\text{mid}] = \text{arr}[3] = 21 < 80$$

$l = \text{mid}$
↓
X (Not a part of answer anymore)

$$l = \text{mid} + 1$$

$$l = 4$$

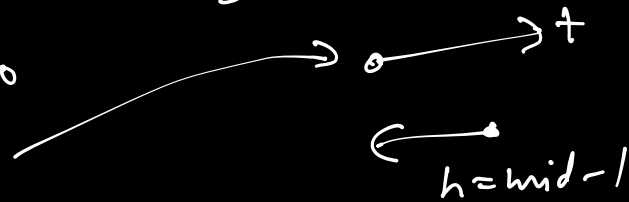
$$l = 4$$

$$h = 7$$

$$\text{mid} = \frac{4+7}{2} = 5$$

$$k = \text{arr}[5] = 40 < 80$$

$$\begin{aligned} l &= \text{mid} + 1 \\ l &= 6 \end{aligned}$$



$$l = 6$$

$$h = 7$$

$$\text{mid} = \frac{6+7}{2} = 6$$

$$k = \text{arr}[6] = 62 < 80$$

$$l = \text{mid} + 1 = 7$$

$$l = 7$$

$$h = 7$$

$$\text{mid} = 7$$

$$k = \text{arr}[7] = 70 < 80$$

$$l = \text{mid} + 1 = 8$$

$$l = 8$$

$$h = 7$$

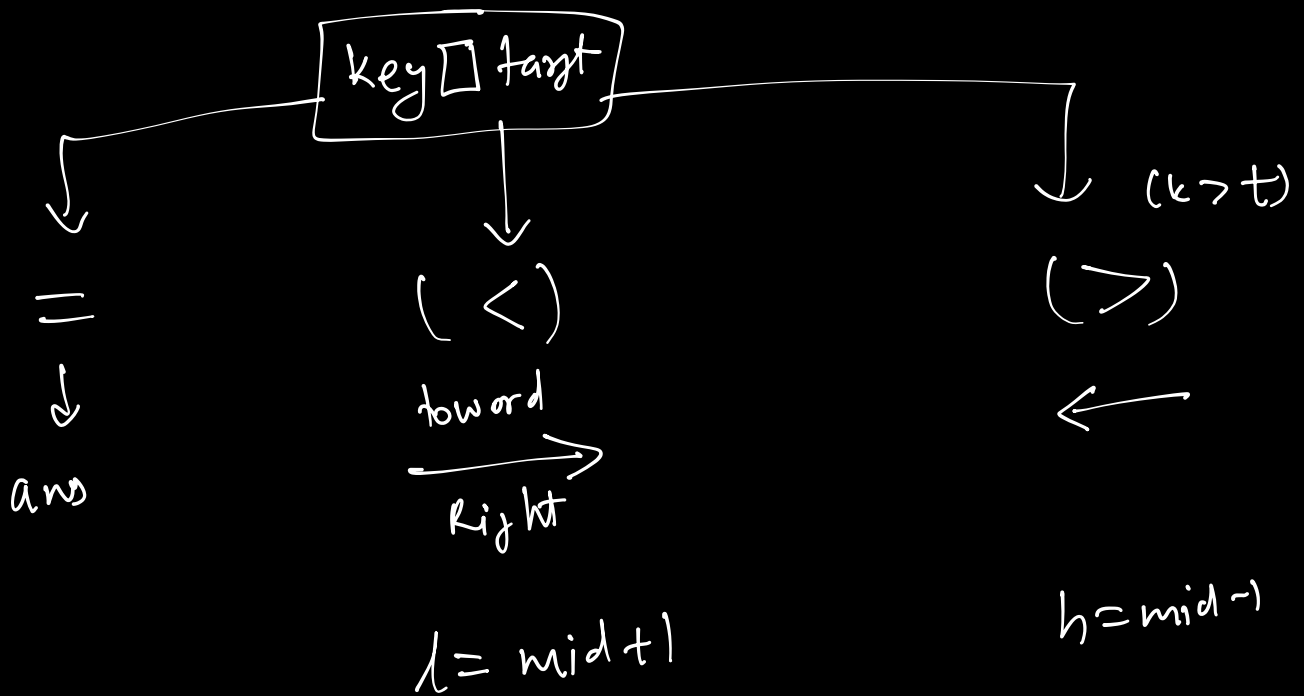
$l > h \rightarrow \text{break}$

Not found
value = 80
ans = -1

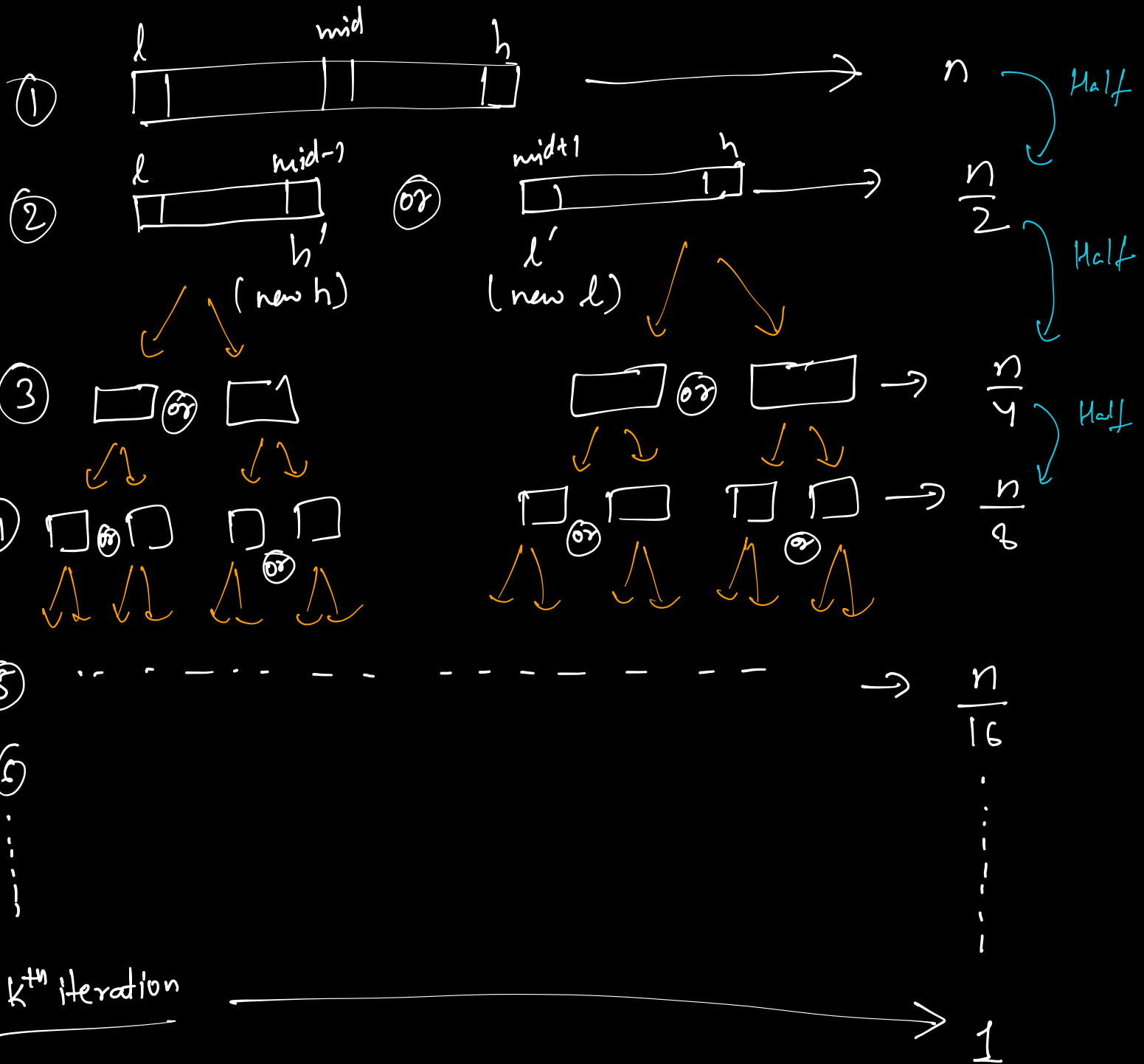
I want to break at $l > h$

working condition

$$l \leq h$$



* How many times we will enter into while loop



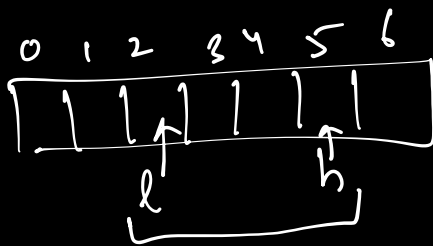
$$k = \log_2 n + 1$$

In Last iteration $\Rightarrow \boxed{l=h} \Rightarrow \underline{1} \quad (h-l+1)$

After

$$\boxed{l > h}$$

$$l=h$$



$$\Rightarrow h-l+1 = 5-2+1 = \underline{\underline{4}}$$

$$n \rightarrow \frac{n}{2} \rightarrow \frac{n}{4} \rightarrow \frac{n}{8} \dots \dots \Rightarrow 1$$

$\frac{n}{2^{k-1}}$

$\uparrow$
 $\nearrow$
 $\searrow$

$$\frac{n}{2^{k-1}} = 1$$

$$n = 2^{\underline{k-1}}$$

$$(k-1) = \log_2 n$$

$$\boxed{k = \log_2 n + 1}$$

$$b = a^x$$

$$x = \log_a b$$

$$\text{fun of TC} = k = \log_2 n + 1$$

$$TC = O(\log_2 n)$$

$$\underline{f_1 = n}$$

$$\underline{f_2 = \log_2 n}$$

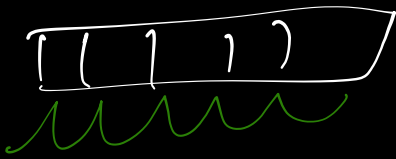
for size of arr = 10000

$$f_1 = 10000 =$$

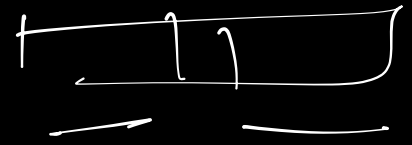
$$f_2 = \log_2(10000) + 1 = 15 =$$

$$\begin{array}{c} n \\ \downarrow \\ 1000000 \end{array} \quad \left| \quad \begin{array}{c} \textcircled{n \log n} + \log n \\ < 1000000 \times 15 \end{array} \quad n = 1000000$$

Linear Search



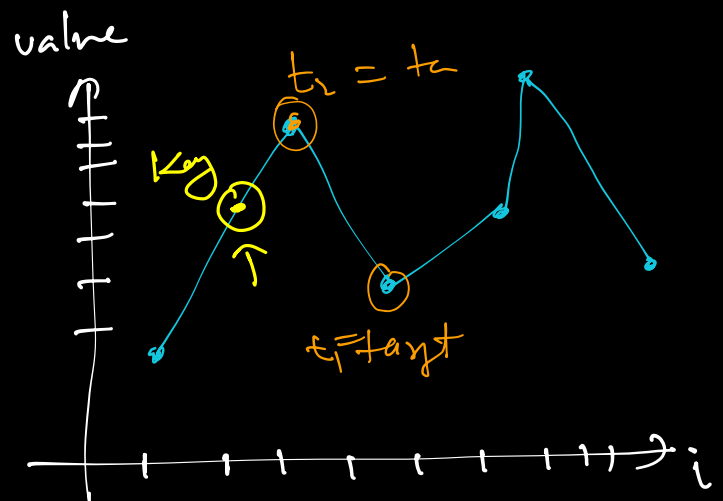
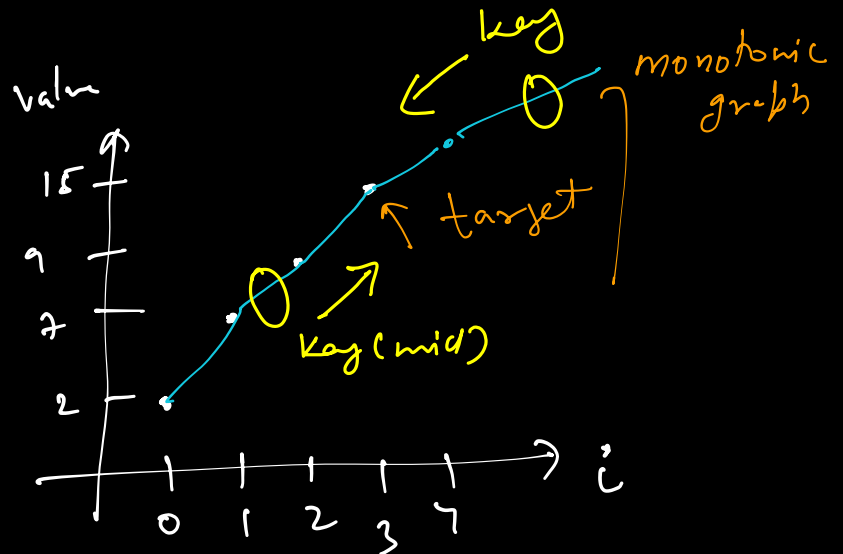
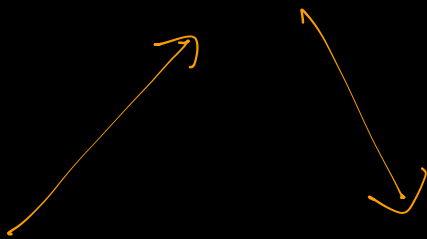
Binary Search

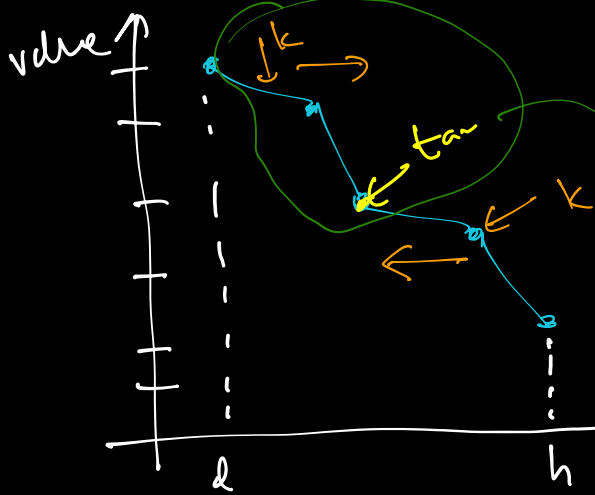


For Binary Search

↳ Sorted Array

0	1	2	3	4
2	7	9	15	20





(Descending order)

l, h

while () {

$$mid = \frac{l+h}{2}$$

$$k = t$$



$$k < t$$

$$h = mid - 1$$

$$k > t$$

$$l = mid + 1$$

150 | 1 | 1 | 1 | 1 | 1

Problem - 1st occurrence

[0 1 2 3 4 5 6 7 8 9 10 11
 [1, 1, 1, 4, 4, 6, 8, 9, 9, 9, 15, 15]
 ↑ ↑ ↺ ↑ ↑ ↑ ↑

$l = 0$ $h = n - 1$
 $= 11$

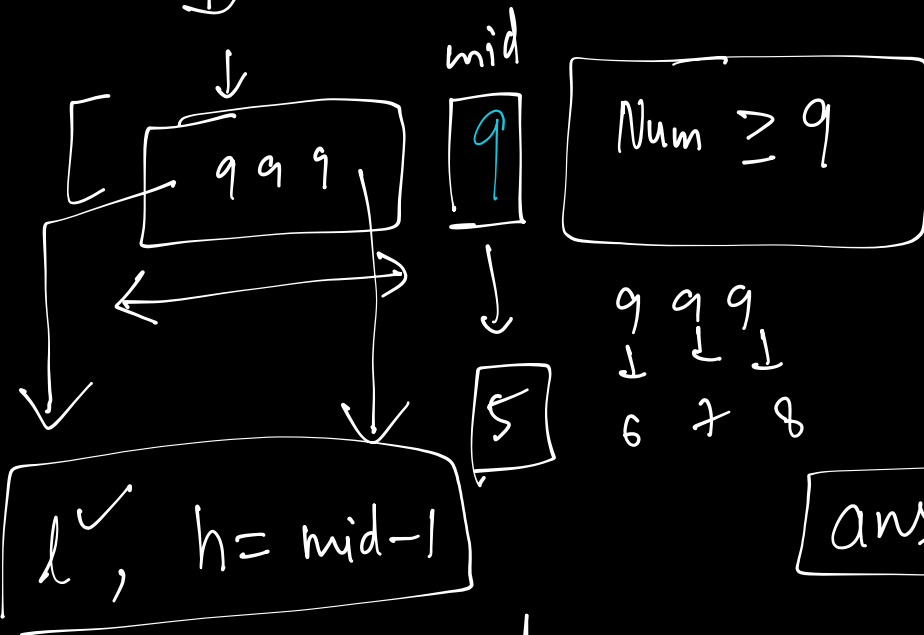
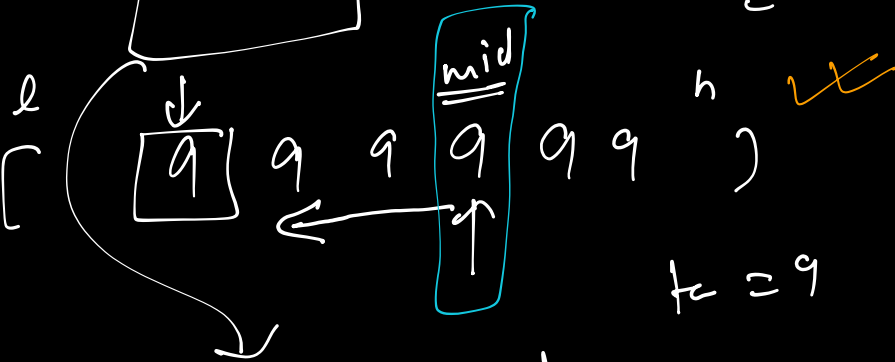
$mid = \frac{11}{2} = 5$

$k = arr[5] = 6$

target = 9

$k < t$

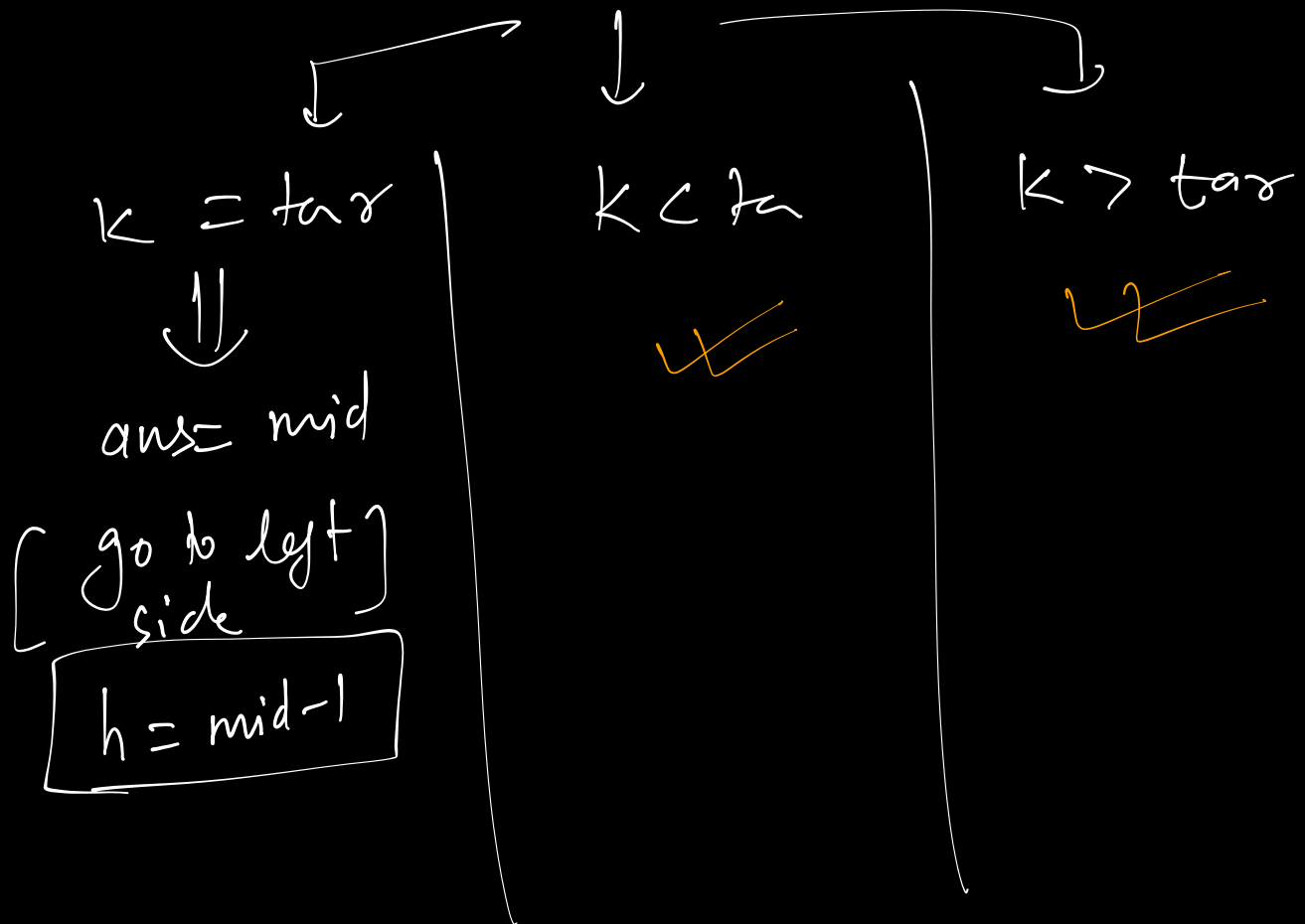
$k > t$



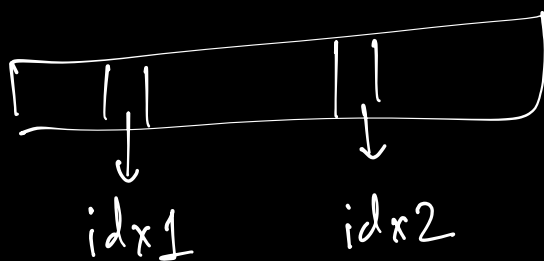
No need to go to right side

ans = mid

[7 8 8 9] → Answer is not present on left
 [7 8 9 9] → " " present on left



No. of Occurrences

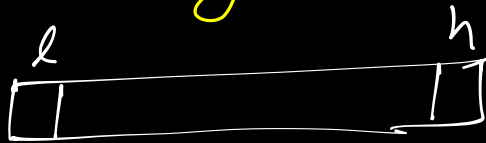


$$freq = idx2 - idx1 + 1$$

if $idx1 = -1$ ←
 $idx2 = -1$ ←
 $ans = 0$

$$ans = freq$$

floor in Array



target < arr[0] $\Rightarrow$ -1 | target > arr[n-1] $\Rightarrow$ arr[n-1]

l = 0 h = n-1

mid = $\left(\frac{l+h}{2}\right)$

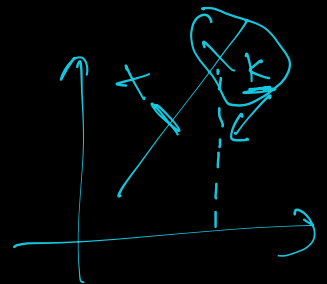


val <= target

$\downarrow$
T / F
1 / 0

0	1	2	3	4
3	6	7	9	15
1	1	1	1	0

target = 14



key target

$k < t$

$q < 14$

ans = mid

go to right

$l = mid + 1$

$k > t$

$\downarrow$

go left

$h = mid - 1$

$\downarrow$

[6] $\rightarrow$ [14]

Implement Ceil